

9.6.6

If n is a positive integer, how many 5-tuples of integers from 1 through n can be formed in which the elements of the 5-tuple are written in decreasing order but are not necessarily distinct? In other words, how many 5-tuples of integers (h, i, j, k, m) are there with $n \geq h \geq i \geq j \geq k \geq m \geq 1$?

The elements of the 5-tuple must be written in decreasing order. Hence, for any five integers selected, there is only one possible ordering.

$$h \geq i \geq j \geq k \geq m \geq 1$$

But repetition is allowed, since the problem states that the elements of the 5-tuple are not necessarily distinct. Therefore, this is an r -combination with repetition allowed.

$$\binom{n+r-1}{r}$$

There are n possible integers, and five integers $h, i, j, k, m \in \mathbb{Z}$ are selected. Hence:

$$\binom{n+5-1}{5} = \frac{(n+4)!}{5!(n+4-5)!} = \boxed{\frac{(n+4)!}{5!(n-1)!}}$$

Solution: There are $\frac{(n+4)!}{5!(n-1)!}$ 5-tuples of integers from 1 through n which can be formed, in which the elements of the 5-tuple are written in decreasing order but are not necessarily distinct.